

PROBLEM SET 4

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions: Due on 25/10/2021; scan or type your work and submit through the link in Moodle. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission.

1. A monopolist can produce a good in different qualities. The cost of producing a unit of quality s is s^3 . Consumers buy at most one unit and have valuation function:

$$v(s, \theta) = \begin{cases} \theta s & \text{if they consume one unit of quality } s, \\ 0 & \text{if they do not consume.} \end{cases}$$

The monopolist decides on the quality (or qualities) it is going to produce and price. Consumers observe qualities and prices and decide which quality to buy if at all. Denote the price (or prices) that the monopolist charges the consumer by P .

- (a) Interpret the parameter θ of the consumers.
 - (b) Characterize the first-best solution, assuming that θ belongs to a subset $\Theta \subset \mathbb{R}$.
 - (c) How does the first-best quality depend on θ ?
 - (d) Now assume that $\Theta = \{\theta_L, \theta_H\}$ with $\theta_L < \theta_H$. Furthermore, suppose that the seller cannot observe θ but believes that $\theta = \theta_H$ with probability β and that $\theta = \theta_L$ with probability $1 - \beta$. Write the optimization program that the monopolist will need to solve.
 - (e) Characterize the second-best solution and the consumer's optimal informational rents.
2. (J. Tirole) Assume that there are two types of consumers for a firm's product, θ_H and θ_L . The proportion of type θ_L consumers is λ . A type θ 's utility when consuming amount x of the good and paying a total T for it is $u(x, T) = \theta v(x) - T$, where

$$v(x) = \frac{1 - (1 - x)^2}{2}.$$

The firm is the sole producer of this good, and its cost of production per unit is $c > 0$.

- (a) Consider a non-discriminating monopolist. Derive his optimal pricing policy, i.e. the cost per unit that the monopolist will charge. Show that he serves both classes of consumers if either θ_L or λ is "large enough." *Hint: Derive the demand curve for each type as a function of the price charged by the monopolist and then compute the "aggregate demand". Use this demand to solve for the optimal price of the monopolist.*
- (b) Consider a monopolist who can distinguish the two types (by some characteristic) but can only charge a simple price p_i for each type θ_i . Characterize the optimal prices. Show that at the optimal prices, the monopolist is charging a marginal price $p = c$ per unit plus a personalized fixed fee equal to the consumer's *net surplus*.
- (c) Suppose that the monopolist cannot distinguish the types. Suppose that the monopolist can charge a single *two-part tariff*. A two-part tariff is an *affine pricing schedule*: $T(x) = A + px$, where A is a fixed fee and p is a marginal price per unit. Derive the optimal two-part tariff under the assumption that the monopolist serves both types.
- (d) Compute the fully optimal nonlinear tariff (second best optimum for the monopolist's screening problem). How do the quantities purchased by the two types compare with the levels obtained in (a)–(c). How do the monopolist's profits from each case in (a)–(d) compare?

3. Let the agent be a drinker who plans to buy at most one bottle of wine within a month. His utility is $U = \theta q - t$ where q is the quality he buys and θ is a positive parameter that indexes his taste for quality. If he decides not to buy any wine, his utility is just 0. Suppose that there are n types of drinkers and that $\theta_1 < \dots < \theta_n$. Their respective prior probabilities are $\pi_1, \dots, \pi_n$ with $\sum_{i=1}^n \pi_i = 1$. The principal is a local monopoly on the wine market. He can produce wine with any quality $q \geq 0$; the production of a bottle of wine with quality q costs him $c(q) = \frac{q^2}{2}$.

- (a) Show that the only binding constraints are the downward adjacent incentive constraints, i.e.

$$\theta_i q_i - t_i \geq \theta_i q_{i-1} - t_{i-1} \quad i = 2, \dots, n$$

and the participation constraint for the lowest type, i.e.

$$\theta_1 q_1 - t_1 \geq 0$$

- (b) Characterize as far as you can the solution to the monopolist's problem.

For the next part of the problem, assume that $n = 3$ and that $\theta_1 = 1, \theta_2 = 5, \theta_3 = 9$.

- (c) Suppose that $\pi_1 = 0.85, \pi_2 = 0.08$ and that $\pi_3 = 0.07$. Compute the optimal qualities for each consumer, the optimal transfers and the monopolist's profits.
- (d) Suppose now that $\pi_1 = \pi_2 = \pi_3 = 1/3$. Compute again the optimal qualities, transfers and the monopolist's profits.
- (e) Compare the results from (c) and (d). What did you find? In which of the two cases is the monopolist performing better? Can you provide an economic interpretation of your results?

For the remainder of the problem, suppose now that the set of types is a continuum, specifically, assume that $\theta \in [\theta_L, \theta_H]$, where $0 < \theta_L < \theta_H$.

- (f) Assume that θ is uniformly distributed on $[\theta_L, \theta_H]$. Characterize the second-best optimal quality-pricing schedule.
- (g) Now suppose that θ is distributed on $[\theta_L, \theta_H]$ according to the distribution $f(\theta) = \frac{2(\theta_H - \theta)}{(\theta_H - \theta_L)^2}$. Characterize the second-best optimal quality-pricing schedule in this case. Is the hazard rate function $\frac{1-F(\theta)}{f(\theta)}$ decreasing? What does a decreasing hazard rate have to do with the monotonicity of the quality-pricing schedule?